Nifty 100 Financial Intelligence Platform

Analyst Guide: screener, dashboard, PDF reports and API

Audience: data analysts who run the pipeline, explore the 92-company universe and share results. No programming is needed to use the dashboard; the command-line steps are copy-and-paste.

Version 1.0. All monetary values are in Indian Rupees, Crore (Rs Cr) unless stated. Stock prices and market caps in the source data are simulated and labelled as such.

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1. Introduction and data caveats

The platform turns raw financial statements for 92 Nifty 100 companies into 50+ KPIs, a screener, peer comparison, valuation flags, cash flow intelligence, auto-generated pros and cons, PDF reports, a Streamlit dashboard and a REST API. Everything is stored in one SQLite file, data/nifty100.db.

What is in the database

Table	Contents
companies, sectors	92 companies, 10 broad sectors with companies, sub-sectors
profitandloss, balancesheet, cashflow	Annual statements, FY2010 to FY2024 (financial year ends in March)
financial_ratios	Computed KPIs per company-year: margins, ROE, ROCE, D/E, ICR, CAGR 3/5/10 years, cash flow KPIs, composite score
capital_allocation	Pattern label per company-year from the signs of operating, investing and financing cash flow
peer_groups, peer_percentiles	11 peer groups and the percentile rank of each member on 10 metrics
market_cap, stock_prices	Valuation multiples and monthly prices. SIMULATED data
documents, prosandcons, pros_cons_generated	Annual report links, source pros/cons, rule-generated pros/cons for all companies

Things to keep in mind

- Market data (prices, market cap, P/E, P/B, EV/EBITDA, dividend yield) is simulated. Use it for screening practice, not for real decisions.
- Latest financial year is 2024-03 for almost every company. SIEMENS reports to September.
- Banks and NBFCs (sector Financials) are exempt from leverage, coverage and margin rules because high D/E is normal for them.
- JIOFIN has only 2 years of history, so it has no tearsheet and several KPIs show N/A.
- Pros and cons are generated from KPI threshold rules with a confidence score. They are screening hints, not investment advice.

2. Setup and daily run order

Open the project folder in VS Code and use the integrated PowerShell terminal. Always run commands from the project root.

One-time setup

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

Every new terminal

```
.\.venv\Scripts\Activate.ps1
```

Rebuild everything from the Excel files (in this order)

```
python -m src.etl.loader               # Excel -> data/nifty100.db (rebuilds it)
python -m src.analytics.run_ratios     # KPIs, capital allocation labels, edge-case log
python -m src.analytics.screener       # output/screener_output.xlsx
python -m src.analytics.peer           # peer percentiles, peer_comparison.xlsx, radar charts
python -m src.analytics.valuation      # valuation_summary.xlsx, valuation_flags.csv
python -m src.nlp.parser               # analysis_parsed.csv, parse_failures.csv
python -m src.nlp.pros_cons_generator  # pros_cons_generated.csv (and database table)
python -m src.analytics.cashflow_intel # cashflow_intelligence.xlsx, distress_alerts.csv
python -m src.analytics.clustering     # cluster_labels.csv, elbow plot, heatmap, outliers, stats
python -m src.reports.build_reports    # tearsheets, sector reports, portfolio summary
```

Warning: the loader deletes and recreates nifty100.db. Run the steps after it again, otherwise the derived tables (ratios, peers, pros and cons) will be missing. If you only changed one analytics module, run that module alone.

Run the applications

```
streamlit run src/dashboard/app.py        # dashboard at http://localhost:8501
uvicorn src.api.main:app --port 8000      # API at http://localhost:8000/docs
pytest tests/ -q                          # test suite
```

The dashboard and the API can run at the same time in two terminals. Stop either one with Ctrl+C.

3. The data pipeline

The pipeline has four layers. Each layer writes files you can open directly.

Layer	What it does	Main outputs
ETL	Reads 12 Excel files, normalises tickers and years, runs 16 data-quality rules, loads SQLite	data/nifty100.db, output/load_audit.csv, output/validation_failures.csv
Ratio engine	Computes KPIs for every company-year with documented edge cases	financial_ratios table, output/capital_allocation.csv, output/ratio_edge_cases.log
Analytics	Screener, peers, valuation, cash flow intelligence, NLP, clustering	output/*.xlsx and *.csv, reports/radar_charts, reports/*.png
Reports and apps	PDF reports, dashboard, REST API	reports/tearsheets, reports/sector, reports/portfolio, src/dashboard, src/api

Reading the data-quality files

- load_audit.csv: rows in, rows out and rejected per table, plus unresolved CRITICAL failures (should be 0).
- validation_failures.csv: one row per rule violation with rule_id, table, company_id, year, field, issue and severity (CRITICAL, WARNING or INFO).
- Rows removed by design: trailing-twelve-month and partial periods, duplicate company-years, and tickers missing from the company master.
- ratio_edge_cases.log: every division by zero, negative equity, debt-free substitution, CAGR turnaround and ROE/ROCE anomaly, with a category.

4. Starting and navigating the dashboard

Start the dashboard with streamlit run src/dashboard/app.py. The sidebar lists the eight screens. Pages load their data from the database and cache it for ten minutes, so a freshly rebuilt database appears after the cache expires or after you restart the app.

Screen	Use it to	Key controls
Home	See market-wide KPIs, the sector mix and the five best companies by quality score	Year selector in the sidebar (2019 to 2024)
Company Profile	Study one company: KPI tiles, 10-year charts, pros and cons	Search box (name or ticker), company dropdown
Screener	Filter all 92 companies with 10 sliders or 6 presets and export CSV	Sidebar sliders, preset buttons, Reset, Download CSV
Peer Comparison	Compare a company with its peer group on a radar chart and a KPI table	Peer group and company dropdowns
Trend Analysis	Overlay up to 3 metrics over 10 years with year-on-year labels	Company picker, metric multi-select
Sector Analysis	See revenue versus ROE per sector and sector median KPIs	Sector dropdown
Capital Allocation	See how companies deploy cash across 8 patterns	Pattern dropdown below the treemap
Annual Reports	Open annual report PDFs per year	Company picker, link-check checkbox

Display conventions

- N/A means the value cannot be computed (zero denominator, negative equity, or too little history). It is never zero.
- Charts are interactive: hover for values, drag to zoom, double-click to reset, use the camera icon to save a PNG.
- Home shows a capped average ROE (5th to 95th percentile) so that a few extreme values do not distort it.

5. Using the Screener

The screener keeps only companies that satisfy every active filter, then sorts them by the composite screen score (0 to 100: profitability 35%, cash quality 30%, growth 20%, leverage 15%).

Step by step

- Open Screener. All 92 companies are listed because every slider starts switched off (at its extreme).
- Click a preset (Quality, Value, Growth, Dividend, Debt-Free, Turnaround) to fill the sliders, or move sliders yourself.
- Read the line above the table: N companies match your filters. The table updates as you move a slider.
- Presets with conditions that have no slider (for example D/E equal to 0) show them under the count as extra filters.
- Press Download CSV to save the table exactly as displayed. Press Reset to clear everything.

Preset definitions

Preset	Conditions	Ranked by
Quality	ROE > 15, D/E < 1, FCF > 0, revenue CAGR 5y > 10	Screen score
Value	P/E < 20, P/B < 3, D/E < 2, dividend yield > 1	FCF yield
Growth	PAT CAGR 5y > 20, revenue CAGR 5y > 15, D/E < 2	PAT CAGR 5y
Dividend	Dividend yield > 2, payout < 80, FCF > 0	Dividend yield
Debt-Free	D/E = 0, ROE > 12, revenue > 5,000 Cr	ROE
Turnaround	Revenue CAGR 3y > 10, FCF > 0, D/E falling	Revenue CAGR 3y

Tips

- Thresholds for the presets live in config/screener_config.yaml. Edit that file to create your own screens, then run python -m src.analytics.screener to refresh the Excel output.
- A company with a missing value (N/A) on a filtered metric is excluded, because it cannot be shown to pass.
- Sliders for D/E, P/E and P/B are maximums; the others are minimums. Interest coverage treats debt-free companies as passing. The D/E maximum is not applied to Financials (banks and NBFCs).
- The same filters are available through the API: GET /api/v1/screener?min_roe=15&max_de=1.

6. Company Profile, Peers and Trends

Company Profile

Type part of a name or ticker in the search box (for example infy), then pick the company. You see six KPI tiles for the latest year, revenue and net profit bars, a dual-axis ROE and ROCE line chart, and pros and cons. Companies with fewer than 10 years show a note with the number of years available.

Peer Comparison

Choose one of the 11 peer groups (for example Private Banks or IT Services) and a company. The radar chart scales each of the 8 axes from 0 to 100 within the group (100 is best; debt-to-equity is inverted). The table below lists every member with the group benchmark highlighted. Companies that belong to no group (about 36 of 92) have no peer view; their tearsheet radar compares them with their sector.

Trend Analysis

Pick up to three metrics. Each gets its own chart because the units differ. The label on every point is the year-on-year percentage change; it is blank when the previous value was zero or negative, since a percentage change would be meaningless.

Reading badges in the peer reports

- Best in Class: top quartile (75th percentile or higher) on at least 6 of 10 metrics within the group.
- Watch List: bottom quartile (25th percentile or lower) on at least 4 of 10 metrics.
- Percentile 1.0 is the best member of the group on that metric; 0.0 is the weakest.

7. Sectors, Capital Allocation and Annual Reports

Sector Analysis

Pick a sector to see a bubble chart (revenue on a log scale versus ROE, bubble size is market cap, colour is sub-sector) and a bar chart comparing the sector's median KPIs with the Nifty 100 median.

Capital Allocation

The label combines the signs of cash flow from operations (CFO), investing (CFI) and financing (CFF) in the latest year.

Pattern	CFO / CFI / CFF	Reading
Reinvestor	+ / - / -	Funds growth and repays debt or pays out from operating cash
Shareholder Returns	+ / - / - with CFO well above profit	Mature cash generator returning cash
Liquidating Assets	+ / + / -	Selling investments or assets while paying down financing
Cash Accumulator	+ / + / +	Raising and generating cash on all fronts
Growth Funded by Debt	+ / - / +	Investing more than it earns, financed by new funding
Distress Signal	- / + / +	Operations burn cash; funded by financing and asset sales
Pre-Revenue	- / - / -	Cash leaving through every channel
Mixed	- / - / + or - / + / -	No clear pattern

Banks often show positive investing cash flow because of how their statements are structured, so read the label for Financials with care.

Annual Reports

Pick a company to list its annual report PDFs from BSE India. Tick the checkbox to test which links still respond; dead links show a red Report unavailable badge. Some links decay over time, which is expected.

8. Generating PDF reports

Three kinds of PDF are generated by one command and saved under reports/.

Report	Location	Contents
Company tearsheet (2 pages)	reports/tearsheets/<TICKER>_tearsheet.pdf	Page 1: KPI tiles, revenue and profit bars, ROE and ROCE chart. Page 2: balance sheet mix, cash flow waterfall, pros and cons, capital allocation badge
Sector report	reports/sector/<SECTOR>_report.pdf	Median KPIs versus the whole universe, best and weakest company, table of every company with 8 metrics
Portfolio summary	reports/portfolio/portfolio_summary.pdf	One page per company in ticker order: 6 KPIs with trend arrows (up = improved, down = declined, right = flat within 2%)

```
python -m src.reports.build_reports --sample   # 5 sample tearsheets only
python -m src.reports.build_reports            # everything (about 40 seconds)
python scripts\sprint5_check.py                # verifies counts and file sizes
```

Rules and checks

- Companies with fewer than 3 years of data are skipped and listed in output/skipped_tearsheets.csv.
- Pros and cons must exist first: run python -m src.nlp.pros_cons_generator before building reports.
- Every tearsheet should be exactly 2 pages and at least 30 KB. The build prints any that are not.
- Open a few PDFs after each full build and check for text overflow or blank pages.

9. Using the REST API

Start the server with uvicorn src.api.main:app --port 8000. Interactive documentation is at http://localhost:8000/docs and the machine-readable specification is docs/openapi.json (import docs/postman_collection.json into Postman). In PowerShell use curl.exe (plain curl is a different command) and keep the quotes around URLs.

Examples

```
curl.exe "http://localhost:8000/api/v1/health"
curl.exe "http://localhost:8000/api/v1/companies?sector=IT"
curl.exe "http://localhost:8000/api/v1/companies/TCS"
curl.exe "http://localhost:8000/api/v1/companies/TCS/pl?from_year=2020-03&to_year=2024-03"
curl.exe "http://localhost:8000/api/v1/companies/TCS/bs"
curl.exe "http://localhost:8000/api/v1/companies/TCS/cashflow"
curl.exe "http://localhost:8000/api/v1/companies/TCS/ratios?year=2024-03"

curl.exe "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1&min_fcf=0"
curl.exe "http://localhost:8000/api/v1/sectors"
curl.exe "http://localhost:8000/api/v1/sectors/IT/companies"
curl.exe "http://localhost:8000/api/v1/peers/IT_Services"
curl.exe "http://localhost:8000/api/v1/companies/TCS/peers/compare"
curl.exe "http://localhost:8000/api/v1/market-cap/TCS"
curl.exe "http://localhost:8000/api/v1/portfolio/stats"
curl.exe "http://localhost:8000/api/v1/companies/TCS/documents"

curl.exe -o TCS.pdf "http://localhost:8000/api/v1/companies/TCS/tearsheet"
```

Endpoint summary (16)

Endpoint	Returns
GET /health	status, row count of every table, uptime, version
GET /companies	All companies; optional sector, market_cap_category, search
GET /companies/{ticker}	Profile plus latest KPIs; 404 for unknown ticker
GET /companies/{ticker}/pl | bs | cashflow	Statement history; optional from_year and to_year (YYYY-MM)
GET /companies/{ticker}/ratios	All KPIs per year; optional year
GET /companies/{ticker}/tearsheet	The tearsheet PDF
GET /screener	Ranked companies; min_roe, max_de, min_fcf, sector, min_rev_cagr_5yr, min_pat_cagr_5yr, max_pe
GET /sectors and /sectors/{sector}/companies	Sector medians; companies of a sector (IT works as an alias)
GET /peers/{group} and /companies/{ticker}/peers/compare	Peer percentiles; radar data versus peer average and benchmark
GET /market-cap/{ticker}	Valuation history (simulated)
GET /portfolio/stats	P10 to P90, mean and standard deviation for 10 KPIs
GET /companies/{ticker}/documents	Annual report links; validate=true checks them live

Status codes

- 200 success. 400 invalid parameter (for example min_roe=abc, or a year not in YYYY-MM format). 404 unknown ticker, sector, peer group or missing tearsheet. 503 database unavailable.
- Responses use null for values that cannot be computed. Every response includes an X-Process-Time-ms header.

10. Analytics outputs: valuation, cash flow, NLP, clustering

File	What it tells you
output/valuation_summary.xlsx, valuation_flags.csv	P/E, P/B, EV/EBITDA, FCF yield and a flag: Caution if P/E is above 1.5 times the sector median, Discount if below 0.7 times, otherwise Fair
output/cashflow_intelligence.xlsx	CFO quality (5-year CFO/PAT), CapEx intensity, FCF CAGR, FCF conversion, distress and deleveraging flags, capital allocation label
output/distress_alerts.csv	Companies with negative operating cash flow and positive financing cash flow in the latest year
output/pattern_changes.csv	Companies whose capital allocation pattern changed from one year to the next
output/pros_cons_generated.csv	12 pro rules and 12 con rules with a 60 to 100 confidence score; sector-relative fallback entries are marked PRO-REL or CON-REL
output/analysis_parsed.csv, cross_validation.csv	Growth numbers parsed from text and checked against the ratio engine (differences over 5 points are flagged DIVERGENT)
output/cluster_labels.csv, cluster_profile.csv	KMeans (5 clusters) on ROE, D/E, revenue CAGR, FCF CAGR and OPM, with archetype names and the companies closest to each centre
reports/elbow_plot.png, correlation_heatmap.png	Why k = 5, and how the 10 core KPIs move together
output/outlier_report.csv	KPIs more than 3 standard deviations from the sector mean (needs sectors of about 10 or more companies)
output/portfolio_stats.csv	Distribution of each KPI across all 92 companies

How to read the cluster names

Names are assigned automatically from each cluster's average profile and are a starting point. If a name does not fit the companies listed in cluster_profile.csv, create config/cluster_names.yaml with lines such as 2: "Value Cyclicals" and run the clustering again.

11. Troubleshooting

Symptom	Likely cause	Fix
ModuleNotFoundError: No module named 'src'	Command run from a subfolder or without -m	cd to the project root and run python -m src.package.module
(.venv) not shown, packages missing	Virtual environment not active	Run .\.venv\Scripts\Activate.ps1; if blocked: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
Dashboard page shows a red error about a missing table	Derived tables were wiped by re-running the loader	Re-run run_ratios, peer, valuation, pros_cons_generator, cashflow_intel
Port 8501 or 8000 already in use	Another dashboard or API is running	Stop it with Ctrl+C, or start with --server.port 8502 (Streamlit) or --port 8001 (uvicorn)
Screener shows 0 companies	Filters too strict, or an N/A value on a filtered metric	Press Reset, then add filters one at a time
Tearsheet missing for a company	Fewer than 3 years of data	Check output/skipped_tearsheets.csv
API returns 404 for a peer group	Name spelled differently	List groups at /api/v1/peers/{any}; spaces and underscores are interchangeable
API screener returns 400	A parameter is not a number or the sector is unknown	Check the message in the response body
Numbers differ from companies.xlsx	Source ROE and ROCE use another definition	The engine value is used for analytics; differences are logged in ratio_edge_cases.log
PDF build says problems (not 2 pages or under 30 KB)	Unusual data for one company	Open that PDF; re-run the build for it and send the ticker to the maintainer
ruff or black reports errors	Style rules, not functional errors	Run black src tests scripts and ruff check src tests scripts --fix

12. KPI reference and glossary

KPI	Formula	Reading
Net profit margin	net profit / sales x 100	Above 10% is good, above 20% excellent
Operating margin (OPM)	operating profit / sales x 100	Cross-checked against the source value
ROE	net profit / (equity capital + reserves) x 100	None when equity is zero or negative
ROCE	EBIT / (equity + reserves + borrowings) x 100	EBIT = operating profit minus depreciation
Debt-to-equity	borrowings / (equity + reserves)	0 means debt-free; above 5 flags non-financials
Interest coverage	(operating profit + other income) / interest	None when interest is 0 (shown as Debt Free); below 1.5 is a warning
Free cash flow	cash from operations + cash from investing	Negative is allowed; three negative years is a concern
CFO / PAT	operating cash flow / net profit	Above 1 high quality, 0.5 to 1 moderate, below 0.5 accrual risk
CapEx intensity	abs(investing cash flow) / sales x 100	Below 3% asset-light, above 8% capital-intensive
CAGR	((end / start) ^ (1 / years) - 1) x 100	None with a flag if the start is zero or either value is negative
Composite quality score	0.30 ROE + 0.25 FCF margin + 0.25 ROCE + 0.20 D/E score	Each part scaled 0 to 100 between the 10th and 90th percentile

Glossary

Term	Meaning
FY	Financial year, April to March. FY24 is labelled 2024-03.
CAGR flags	TURNAROUND (loss to profit), DECLINE_TO_LOSS, BOTH_NEGATIVE, ZERO_BASE, INSUFFICIENT (too little history).
Debt-free	Borrowings equal to zero. D/E is 0 and interest coverage is displayed as Debt Free.
Percentile rank	Position within a peer group from 0 (weakest) to 1 (best).
Winsorise	Clip values to the 5th to 95th (or 10th to 90th) percentile before scaling.
Tearsheet	Two-page company summary PDF.